

Question Paper

Exam Date & Time: 22-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH. EXAMINATIONS - NOVEMBER 2025

Quantum Computing [CSS_5104]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) What is entanglement? Briefly explain the challenges associated with the Quantum computers. (2)

A) Course Outcome: "CO1" / Cognitive Level: "Remember"

- B) Verify if the matrix C given below is Unitary or not? Find its determinant. (4)

$$C = \begin{bmatrix} 1-i & 2 & \sqrt{1+i} \\ -i & 1+2i & 2 \\ 3i & \sqrt{2}+i & -1 \end{bmatrix}$$

Course Outcome: "CO1" / Cognitive Level: "Understand"

- C) (4)

Consider the quantum state $|\Psi_1\rangle = \frac{1}{\sqrt{5}} (|00\rangle + i|01\rangle - \sqrt{2}|10\rangle + |11\rangle)$.

The second qubit is measured. What is the probability that it is 1? What is the probability that it is 0? Write down the normalized post measurement states for each case. Suppose now the first qubit is measured. What is the probability that it is 0? What is the probability that it is 1?

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2) Explain the four postulates of Quantum mechanics. (4)

A) Course Outcome: "CO1" / Cognitive Level: "Remember"

- B) Given a quantum state $|\Psi\rangle = \frac{1}{\sqrt{4}} (|0\rangle + i\sqrt{3}|1\rangle)$, a sequence of X, Z, and Y gates are applied on this state. Show the intermediate outputs and the final output. (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- C) Write the quantum circuit to create the four Bell States and explain. What are the applications of Bell States? (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

3) What is Quantum Fourier Transform? Write the associated quantum circuit and derive the expression for Quantum Fourier Transform. (5)

A)

Course Outcome: "CO3" / Cognitive Level: "Understand"

B) Explain how Deutsch-Jozsa algorithm is used to test if a given function is constant or balanced. (5)

Course Outcome: "CO4" / Cognitive Level: "Understand"

4) Design a Quantum Circuit for Bernstein-Vazirani algorithm and explain how it uses just one query to find c , where (4)

A)

$f(x) = c.x$

Course Outcome: "CO4" / Cognitive Level: "Application"

B) Give an example and explain B92 protocol. (3)

Course Outcome: "CO3" / Cognitive Level: "Remember"

C) Explain single qubit Phase Error Detection and Correction using a 3-Qubit System. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

5) Draw the Quantum circuit and explain the working of Quantum Teleportation protocol. (4)

A)

Course Outcome: "CO3" / Cognitive Level: "Remember"

B) Write short notes on the following: (6)

- i) Quantum Parallelism and Entanglement
- ii) Quantum Key Distribution

Course Outcome: "CO3" / Cognitive Level: "Remember"

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Event Name	Subject
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058-END SEMSTER-ODD-2025-PG CSS_5104:Quantum Computing

#	Course Outcome	Description	Marks
1	1	CO 1	14
2	2	CO 2	6
3	3	CO 3	18
4	4	CO 4	9
5	5	CO 5	3
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025-PG CSS_5104:Quantum Computing

#	Level	No of Questions	Marks (%)
1	Remember	5	19(38%)
2	Understand	7	27(54%)
3	Application	1	4(8%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)

Total	13	50 (100%)
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